DANIEL BENJAMIN

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available starting May 2026 for a 4-month or 8-month co-op position

EDUCATION

University of British Columbia Expected Graduation: May 2028
Bachelor of Applied Science – Mechanical Engineering

SUMMARY

Mechanical Engineering student with experience supporting building systems design, project coordination, and document control processes. Proven ability to improve workflow efficiency (30%+ time reduction) and manage technical documentation across multidisciplinary teams. Strong attention to detail with experience coordinating stakeholders, maintaining version-controlled records, and supporting project delivery under tight deadlines.

SELECTED EXPERIENCE

Mechanical Co-op, Avalon Mechanical May 2025 – Aug 2025

- Developed AutoLISP automation scripts reducing CAD drafting time by over 30%, saving 7+ hours per week
- Assisted with drawing revisions, redlines, and documentation for tender and construction packages
- Performed hallway pressurization analysis and fan sizing using Excel and TRACE, calculating required airflow and pressure differentials to meet NFPA and provincial code requirements
- Validated fire sprinkler hydraulic systems using Elite Software FIRE for code-compliant designs

Chassis Team Lead (prev. Team Member)**,** UBC Formula Electric Sep 2023 – Present

- Led 7-person chassis subteam using agile project management practices, tracking tasks and milestones in Jira and maintaining Gantt charts to keep the 2026 competition vehicle on schedule
- Implemented a CAD document control system using SVN on Oracle Cloud to track and manage design revisions, paired with a custom SOLIDWORKS integration to streamline adoption
- Coordinated with other subteams to ensure chassis design met packaging and schedule requirements
- Wrote detailed design documentation and SOPs to ensure knowledge transfer for future team members
- Designed a custom 3D printer management dashboard and Slack integration to monitor print status and automate notifications, manage reservations and reduce downtime of 3D printers

Undergraduate Research Assistant, UBC MEMS Lab May 2024 – Feb 2025

- Systematically reviewed and documented 15+ prosthetic hand designs, compiling findings into a structured format to guide the team's design direction
- Rapidly prototyped design iterations using a custom-built multi-extrusion 3D printer running RepRapFirmware
- Helped develop a demonstration platform to allow the prosthetic hand to mimic human hand gestures using MATLAB and Arduino scripts along with a Leap Motion hand-tracking camera module

Volunteer, Victoria Hand Project July 2022 – Present

- Assembled and repaired 3D-printed low-cost prosthetic arms with voluntary open/close functionality
- Identified process improvements and contributed to creation of new assembly documentation to improve quality and consistency

SVNWorks, Personal Project danielrbenjamin.com/projects/svnworks

- Built a custom integration between SOLIDWORKS, File Explorer, and an SVNserver to enable version-controlled document management for a student engineering team
- Automated check-in/check-out workflows and revision tracking, replacing manual file management with a structured document control system

SKILLS

Mechanical: SOLIDWORKS, AutoCAD (+ AutoLISP), ANSYS, Composites, Machining, 3D Printing, FMEA
Software: MS Office (Excel, Project), C, MATLAB, Python, JS, Git, SVN, Jira, Bluebeam, Docker, Adobe CC